A PROJECT REPORT ON

SOCIAL MEDIA FOLLOWERS PREDICTION

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GOVERNMENT POLYTECHNIC FOR WOMEN

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INTRODUCTION:

There are so many social media platforms today where you will find so many content creators in so many types of fields. As a social media consumer, the number of followers you have may not be of interest to you, but as a content creator or as a businessman, the number of followers you have is important for your content for reaching more audience. So, the task of social media followers prediction is very valuable for every content creator and every business that relies on social media. So if you want to learn how to predict your social media followers for the next month, this article is for you. In this article, I will walk you through the task of social media followers prediction with machine learning using Python.

Social Media Followers Prediction:

To predict the increase in the number of followers you can expect to see, you need a data set of your social media followers that can show you the activities of people in your social media account like:

1. how many people have followed you every month
2. how many views results in how many followers
3. how many of your followers unfollow you every month

So it is very difficult to find such a datasets among the most common social media platforms like Facebook and Instagram as these platforms do not provide any data related to your followers. So for the task of social media followers prediction with machine learning, I collected data from my social media account on Medium, which is a social media platform for content writers, bloggers, and researchers. You can use the same process on your datasets whether you get it from Medium, Instagram, or any other social media application to predict your social media followers. For practice, you can use the same dataset that I am using.

DATASETS: Ten real-valued features are computed for each cell nucleus:

- Period Start
- Period End
- Followers_gained
- Followers_lost
- Followers_net
- Followers_total
- Subscribers_gained
- Subscribers_lost
- Subscribers_net
- Subscribers_total
- Views

Social Media Followers Prediction using Python:

I will start the task of social media followers prediction with machine learning by importing the necessary Python libraries and the dataset that I have collected about my followers from Medium:

Code:

```
import pandas as pd

import matplotlib.pyplot as plt

import seaborn as sns

import numpy as np

from autots import AutoTS

data = pd.read_csv('C:/Users/91935/OneDrive/Desktop/kush.csv')

data.drop(data.tail(1).index, inplace=True)

data.head()
```

Output:

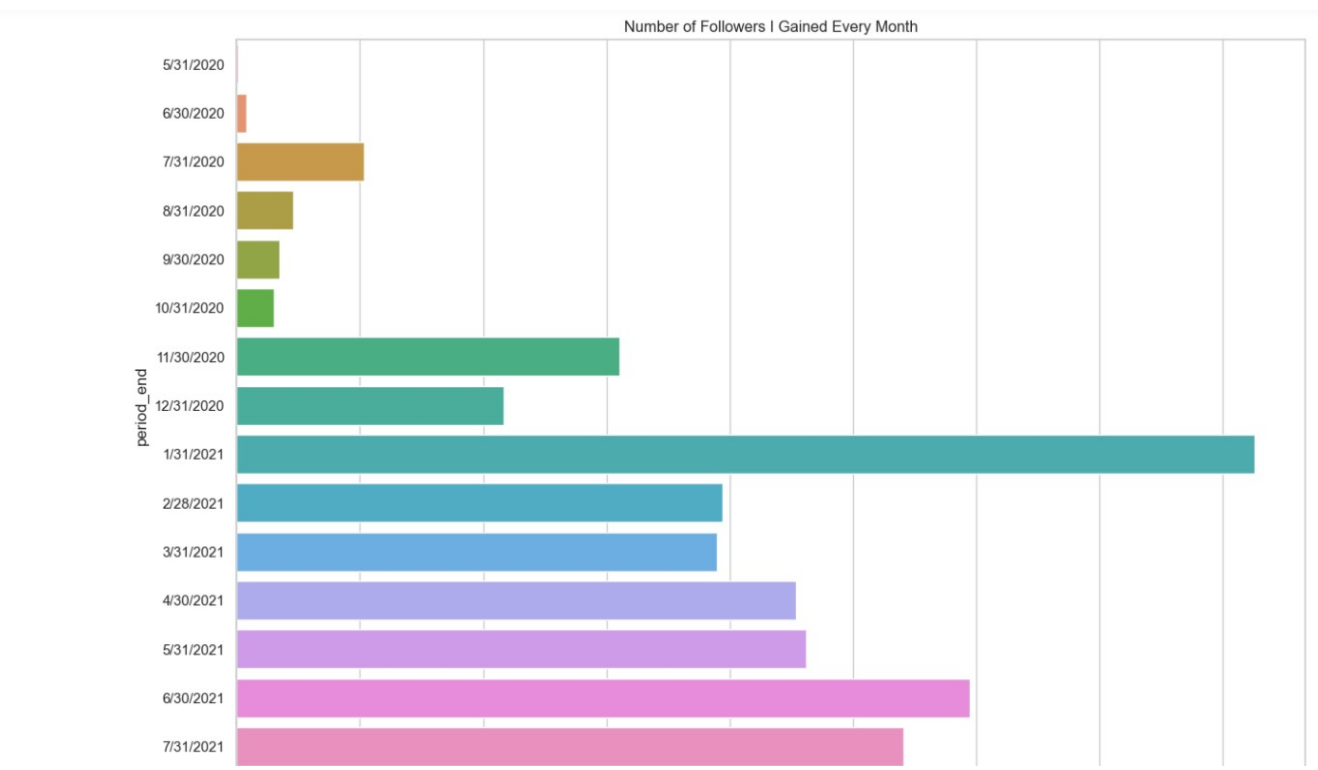
	period_start	period_end	followers_gained	followers_lost	followers_net	followers_total	subscribers_gained	subscribers_lost	subscribers_net	subscribers_total	views
0	05-01-2020	5/31/2020	1	0	1	1	0	0	0	0	128
1	06-01-2020	6/30/2020	8	0	8	9	0	0	0	0	16130
2	07-01-2020	7/31/2020	103	0	103	112	0	0	0	0	14616
3	08-01-2020	8/31/2020	46	0	46	158	0	0	0	0	4053
4	09-01-2020	9/30/2020	35	1	34	192	0	0	0	0	5153

Now I will have a look at the number of followers that I gained every month on my account since I joined this social media platform:

Code:

```
plt.figure(figsize=(15, 10))
sns.set_theme(style="whitegrid")
plt.title("Number of Followers I Gained Every Month")
sns.barplot(x="followers_gained", y="period_end", data=data)
plt.show()
```

Output:

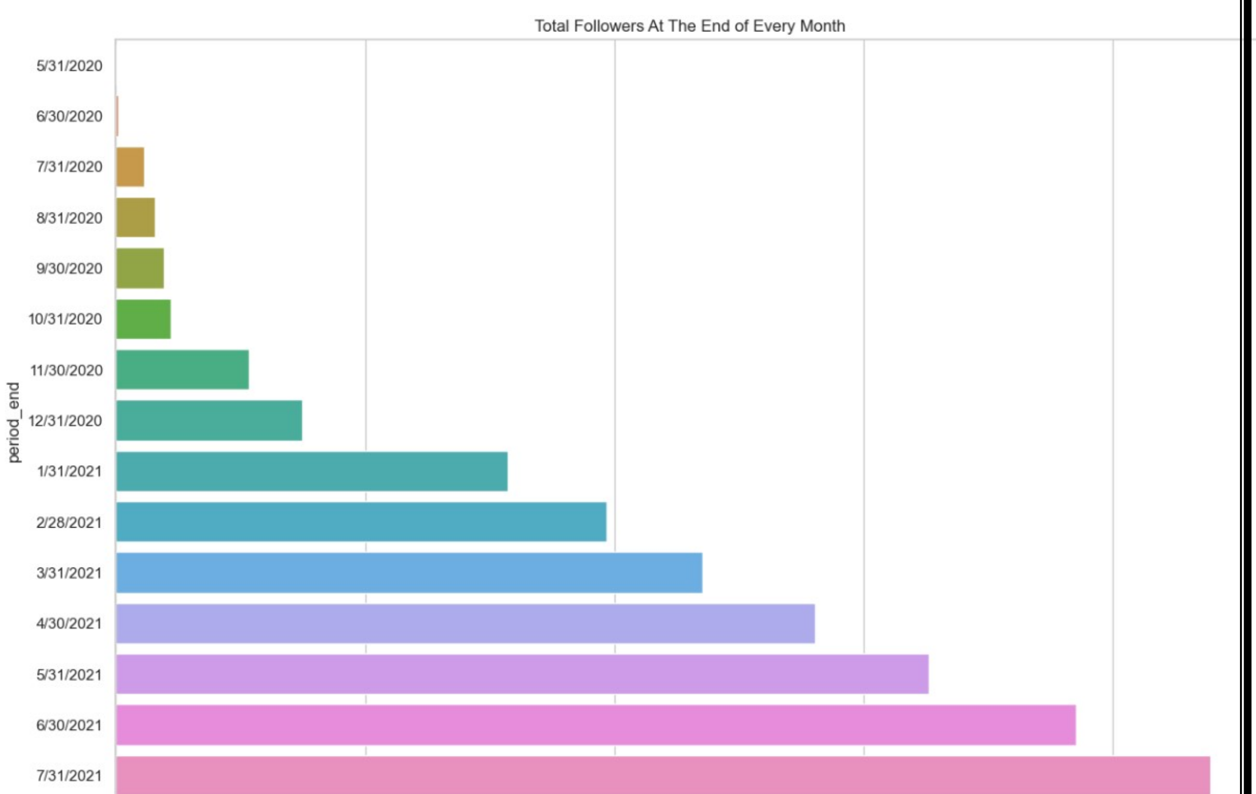


Now let’s have a look at the total number of followers I end up with every month:

Code:

```
plt.figure(figsize=(15, 10))
sns.set_theme(style="whitegrid")
plt.title("Total Followers At The End of Every Month")
sns.barplot(x="followers_total", y="period_end", data=data)
plt.show()
```

Output:

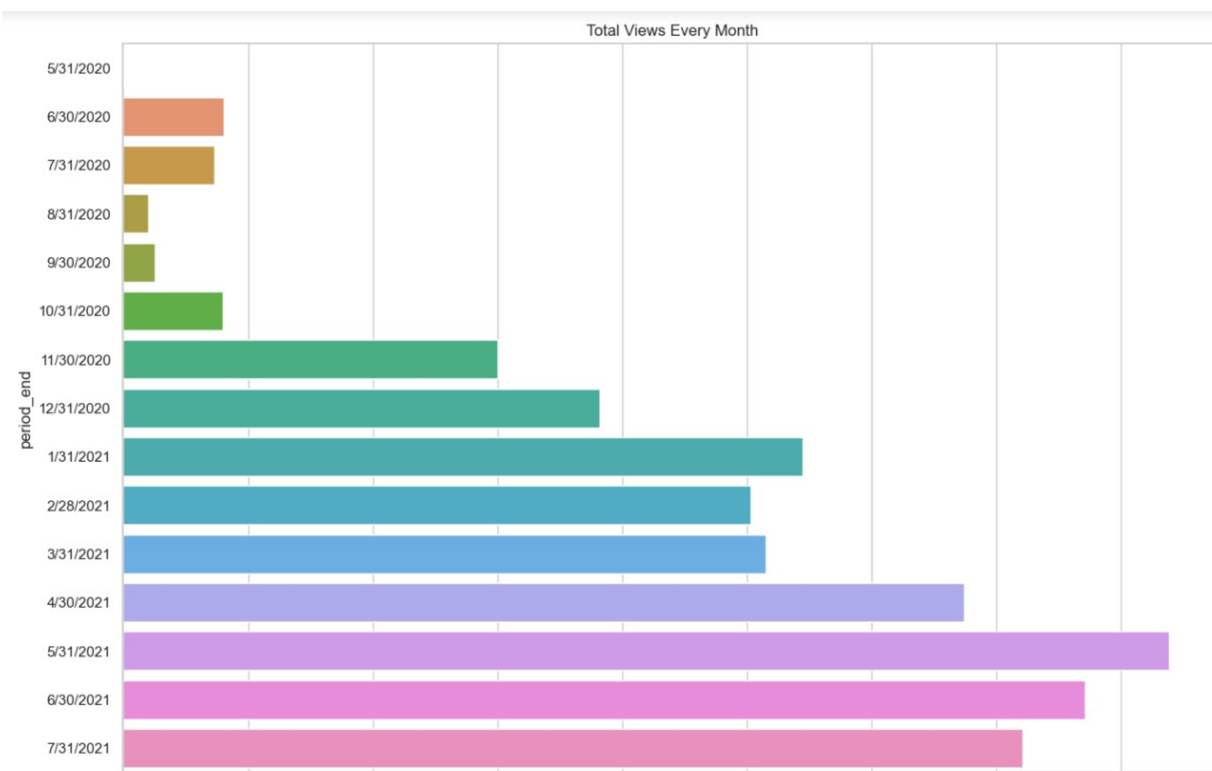


Now let’s take a look at one of the most important features, which is the total number of views I get each month:

Code:

```
plt.figure(figsize=(15, 10))
sns.set_theme(style="whitegrid")
plt.title("Total Views Every Month")
sns.barplot(x="views", y="period_end", data=data)
plt.show()
```

Output:



Now I will be using the autots library in Python, which is one of the best data science libraries for time series forecasting. If you have never used this library before, you can easily install it on your system using the pip command:

Code:

```
from autots import AutoTS

model = AutoTS(forecast_length=4, frequency='infer',
ensemble='simple')

model = model.fit(data, date_col='period_end',
value_col='followers_gained', id_col=None)

prediction = model.predict()

forecast = prediction.forecast

print(forecast)
```

Output:

```
Using 3 cpus for n_jobs.
Data frequency is: ME, used frequency is: ME
Model Number: 1 with model ARIMA in generation 0 of 20
Model Number: 2 with model AverageValueNaive in generation 0 of 20
Model Number: 3 with model AverageValueNaive in generation 0 of 20
Model Number: 4 with model AverageValueNaive in generation 0 of 20
Model Number: 5 with model DatepartRegression in generation 0 of 20
Model Number: 6 with model DatepartRegression in generation 0 of 20
Model Number: 7 with model DatepartRegression in generation 0 of 20

C:\Users\91935\anaconda3\Lib\site-packages\sklearn\svm\_classes.py:32: FutureWarning: The default value of `dual` will change f
rom `True` to `'auto'` in 1.5. Set the value of `dual` explicitly to suppress the warning.
  warnings.warn(
C:\Users\91935\anaconda3\Lib\site-packages\sklearn\svm\_base.py:1242: ConvergenceWarning: Liblinear failed to converge, increas
e the number of iterations.
  warnings.warn(

Model Number: 8 with model DatepartRegression in generation 0 of 20
Epoch 1/50
1/1 [==============================] - 5s 5s/step - loss: 0.3676
Epoch 2/50
1/1 [==============================] - 0s 7ms/step - loss: 0.3768
Epoch 3/50
1/1 [==============================] - 0s 6ms/step - loss: 0.3675
Epoch 4/50
1/1 [==============================] - 0s 7ms/step - loss: 0.3665
Epoch 5/50
1/1 [==============================] - 0s 6ms/step - loss: 0.3587
Epoch 6/50
1/1 [==============================] - 0s 7ms/step - loss: 0.3633
Epoch 7/50
1/1 [==============================] - 0s 7ms/step - loss: 0.3659
Epoch 8/50
1/1 [==============================] - 0s 7ms/step - loss: 0.3596
Epoch 9/50
1/1 [==============================] - 0s 7ms/step - loss: 0.3549
Epoch 10/50
```

So this is how you can predict the increase in the number of your followers on any social media platform. As a social media consumer, the number of followers you have may not be of interest to you, but as a content creator or as a businessman, the number of followers you have is important for your content for reaching more audience. I hope you liked this article on the task of social media followers prediction with machine learning using Python. Feel free to ask your valuable questions in the comments section below.

ABSTRACT:

Social media is popular in society right now. People are using so many social media platforms today where you will find so many content creators. As a social media consumer, the number of followers you have may not be of interest to you, but as a content creator or as a businessman, the number of followers you have is important for your content for reaching more audience. So, the task of social media follower prediction is very valuable for every content creator and every business that relies on social media. The social media follower prediction system is a machine learning-based solution designed to predict the number of followers for a social media account in the future based on historical data. The system takes into account various factors such as engagement rate, post frequency, and time to make accurate predictions. The predictions made by the system can be used to optimize social media strategies and to identify areas that require improvement.

PROBLEM STATEMENT:

The Social Media Followers Prediction System is to develop a machine learning-based solution that accurately predicts the number of followers for a social media account in the future based on historical data. The system must consider various parameters such as engagement rate, post frequency, and time to make accurate predictions. The system should also be adaptable to changes in the social media platform's algorithm, which can affect the engagement rate and the number of followers. The system must provide actionable insights to businesses and content creators to optimize their social media strategy and achieve their goals.

IMPLEMENTATION:

To build a Social Media Follower Prediction System it’s necessary to install the following libraries pandas, sklearn, matplotlib using the import method. Linear regression, r2_score, plt are imported as well. Then we load the CSV file and processor the data. The feature and model selection were done, followed by the model is been trained. Once after the training phase is completed, the model is been evaluated and deployed. Then the system is ready to predict the followers in the future.

Solution:

Prediction 1: Authenticity will be more important than ever

Prediction 2: Creators and influence's, and knowing how to work with them, will continue to be important

Prediction 3: There will be a renewed focus on social media customer service

Prediction 4: Brands will use a variety of content types

Prediction 5: Data usage will become more sophisticated and cross-functional

Prediction 6: Optimizing existing platforms

Prediction 7: The future of AI in social media will remain top-of-mind

Benefits of using a social media management workflow:

- More time for creativity
- More quality control checkpoints
- Stronger accountability and collaboration
- Easier on boarding and adaptability
- Easily manage campaign results

FUTURE SCOPE:

The future scope of the Social Media Followers Prediction System is very promising. With the ever-growing popularity of social media platforms and the increasing number of content creators and businesses relying on social media for marketing, this system will continue to be in high demand. Some of the potential future advancements in this system include incorporating more advanced machine learning algorithms such as deep learning, integrating more data sources such as user demographics and behaviors, and expanding the system to work with multiple social media platforms. Additionally, this system could also be used for other purposes such as predicting the popularity of a certain type of content or product, which could be useful for businesses in product development and marketing. Overall, the future of the Social Media Followers Prediction System is bright and has the potential to be an important tool for content creators and businesses in the digital age.

CONCLUSION:

In conclusion, the Social Media Followers Prediction System is a useful tool for content creators and businesses to estimate their future number of followers on social media platforms. The system leverages machine learning algorithms and historical data to make accurate predictions based on various factors such as engagement rate, post frequency, and time. The use of this system can aid in the planning of social media strategies and help content creators and businesses reach their target audience more effectively. However, it is important to note that the accuracy of the predictions may vary depending on the quality and quantity of data available, as well as changes in social media algorithms and user's behavior. Therefore, it is recommended to periodically review and update the system to ensure optimal performance. Overall, the Social Media Followers Prediction System has the potential to provide valuable insights and help content creators and businesses succeed in the competitive world of social media.